Revolutionising Lithium-Ion Battery Manufacturing using Hydrophobin-based Proteins, Generating $250M of Revenue Within a Decade

TKS x MAVERICK BIOMETALS

A New Way to Manufacture Lithium-Ion Batteries — Powered by Proteins

The Problem

Toxic PVDF binders and NMP solvents make battery manufacturing unsafe, expensive, and unsustainable.

➜ Lithium-ion battery factories still rely on PVDF binders and NMP solvents, which require costly infrastructure like explosion-proof ovens, special ventilation, and solvent recovery systems.

➜ This adds up to $30M+ in CAPEX per GWh, with additional OPEX due to long drying times and toxic emissions exceeding the 5 mg/m^3 safety limit by up to 10×.

➜ These inefficiencies stagnate factory ROI, increase regulatory risks under TSCA and EU REACH, and expose workers to long-term health hazards.

The Solution

AquaFuse: Protein-powered enhancement of water-based binders for faster drying and seamless integration.

➜ AquaFuse, a Class II hydrophobin-based protein, integrates with existing SBR/CMC water-based binders to reduce drying time from 10–15 min to 6–8 min, slashing OPEX by up to 60% per factory.

➜ No changes to machinery — factories can simply add AquaFuse to their binder slurry and run trials on a single line before scaling up, ensuring full performance transparency.

➜ Partnering with battery manufacturers to co-develop and test the formulation, while Maverick Biometals handles protein manufacturing and licensing to ensure scalability and IP protection.

The Outcome

Improved margins, safer factories, and $60M+ savings per plant over 3 years.

➜ Enables $60M in operational cost savings per factory over 3 years, with just a $4.5M investment, yielding an ROI of >12×.

➜ Removes up to 500,000 kg of NMP vapor per site and 1800 tons of CO_2, dramatically improving factory sustainability while meeting upcoming TSCA restrictions.

➜ Unlocks scalable licensing and subscription revenue for Maverick, while empowering manufacturers to meet global sustainability, cost, and safety mandates with a single formulation shift.

Binder Technology in Batteries Has a Rising Demand

Binders in Batteries: Polymers hold electrode materials together and ensure adhesion to current collectors.
Current Standard: Most lithium-ion batteries use PVDF (Polyvinylidene fluoride) with NMP (N-methyl-2-pyrrolidone) as a solvent for strong adhesion.
Rising Demand: EV battery demand is **expected to grow 7× by 2030**, increasing PVDF and NMP use.
Challenges: More batteries mean **higher emissions, hazardous waste, and rising production costs.**

***Refer to Appendix A for relevant calculations and justification.**

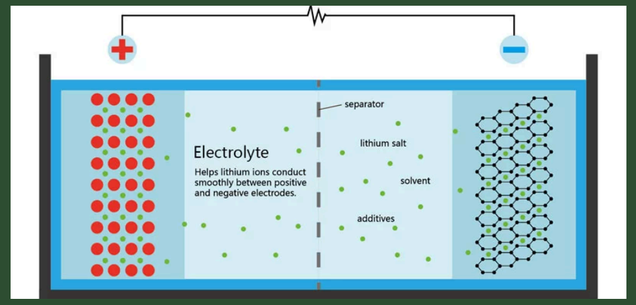
Key problems include:



Highly toxic substances used in current binder technology, leading to fines, and very difficult to dispose – hazardous waste.



Slow-drying binders like PVDF create bottlenecks in battery production. They reduce manufacturing speed, increase energy use, and limit scalability.



Low electrode mechanical stability & adhesion due to weak alternative binders, affecting battery lifespan drastically

NMP Solvent is the Expensive Poison in Battery Manufacturing

$50k

in fines per day for using NMP, set by the U.S Environmental Protection Agency. If a plant gets audited and lacks proper NMP control, it can be **shut down immediately**.

Upto 50 mg/m^3

of NMP vapor is produced during Lithium-Ion battery manufacturing—**10× the safe limit of 5 mg/m^3**. Most plants report 10–30 mg/m^3 during coating, but poorly regulated sites exceed 50 mg/m^3

$30M

is needed for a 1 GWh battery plant using NMP/PVDF, due to **explosion proofing, solvent recovery, and ventilation.**

75-80%

of lithium-ion batteries use PVDF binders with NMP, putting most EV battery plants at **risk of regulatory violations by 2025–2026.**

***Refer to Appendix A for relevant calculations and justification.**

AquaFuse

A less toxic and more cost-effective solution!

AquaFuse's core innovation is water-based binders, offering a safer and more efficient alternative:

$30M

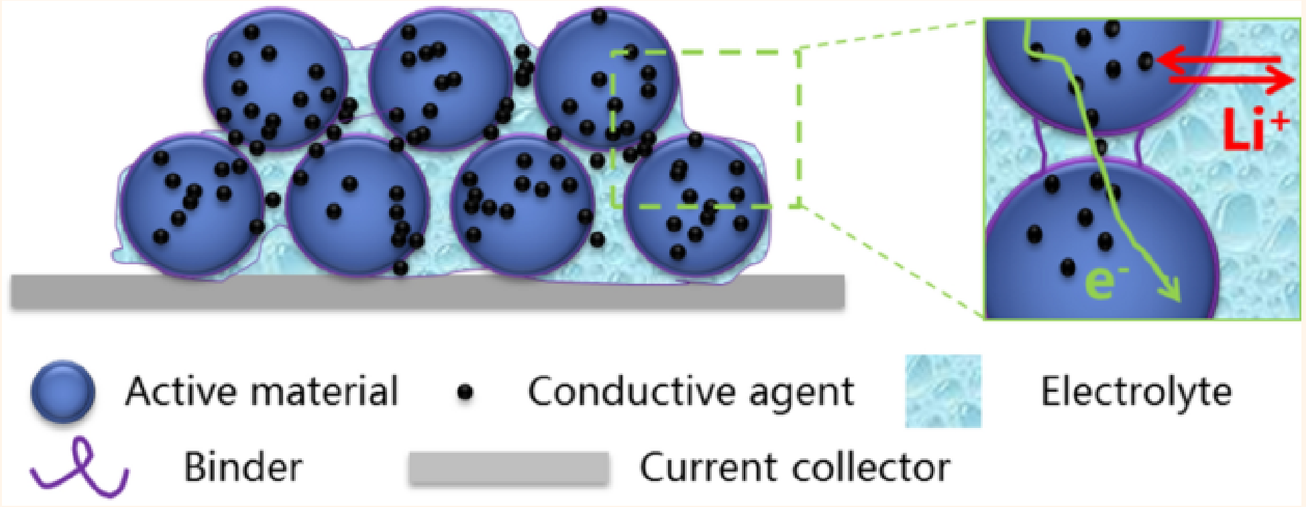
in upfront CAPEX savings per GWh by eliminating the need for **solvent recovery, explosion-proofing, and high-end ventilation systems** required for PVDF/NMP handling.

100%

non-toxic water-based solvent replaces hazardous NMP, drastically **reducing factory emissions, VOC exposure, and compliance risks** — enabling safer work environments and greener supply chains.

***Refer to Appendix B for relevant calculations and justification.**

Water-based binder technology is used due to its hydrogen bonds:



Use

Water-based binders enable safer, eco-friendly lithium-ion battery manufacturing by holding materials together **without toxic solvents like NMP.**

Properties

Water-based binders, like CMC-SBR, form strong hydrogen bonds, **improving adhesion and stability**. They enable electrode processing at 25–60°C vs. PVDF's 150–200°C**.**

***Refer to Appendix B for relevant calculations and justification.**

1/3rd

reduction in throughput

A 20 GWh/year factory could drop to **~13 GWh/year** without added lines or longer ovens, making drying a key factor.

30-50%

higher energy use per kWh → higher OPEX cost

Longer drying times increase electricity use for heating and airflow, **raising operational costs** and straining factory energy systems.

***Refer to Appendix B for relevant calculations and justification.**

However, Long drying times of water-based binders cause immense roadblocks.

Water has a high heat of vaporization, so removing it from thick electrode coatings can take longer, potentially slowing the coating line.

Committed companies are highly affected:

LG Energy Solution, one of the world's top battery manufacturers, has committed to reducing reliance on PVDF and NMP by shifting to water-based systems across select facilities. However, its transition has stalled due to slow drying times, limiting full-scale deployment and cost gains.

LG is exploring CMC-SBR water-based binders to replace toxic PVDF-NMP systems. While early trials reduce solvent risk, they face manufacturing inefficiencies due to **10–15 min drying bottlenecks**.

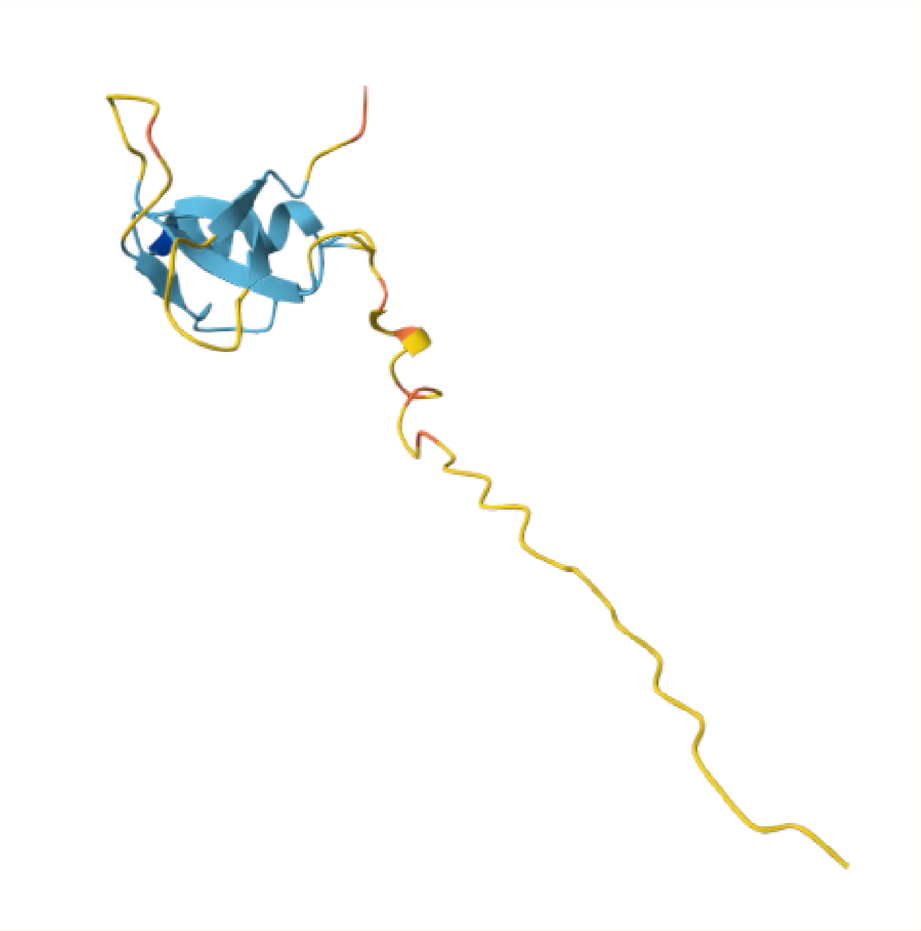
Without a solution to enhance drying speed, LG's transition remains limited to small-scale lines — **hindering broader rollout**, cost savings, and sustainability targets. This is where AquaFuse can deliver breakthrough impact.

***Refer to Appendix B for relevant calculations and justification.**

Part 2 of AquaFuse:

Hydrophobin Class II proteins for improving drying times of Water-Based Battery Binders

Hydrophobin-enhanced binders enable faster drying through surface tension modulation



Use

Hydrophobin-enhanced water-based binders repel excess moisture during drying, thanks to their ability to form hydrophobic protein films at interfaces. This promotes **faster evaporation, smoother coatings, and less energy-intensive drying.**

Properties

Hydrophobins are amphiphilic proteins that migrate to the slurry surface, forming a structured layer that repels water, reduces retention, and prevents cracking. This enables **uniform drying in just 6–8 minutes**, down from 10–15 minutes with standard water-based systems.

***Refer to Appendix B for relevant calculations and justification.**

All available evidence supports that what we're doing is grounded in reality and likely to succeed.

(Published Data, Preliminary Tests, Partner Interest)

Our cost modeling, based on industry data, projects substantial savings. Our protein-enhanced binder offers a clear advantage by **reducing drying time and energy use** compared to standard water-based binders.

$15 savings per kWh

1.85x Greater Production Output

40% Reduced Energy Usage

***Refer to Appendix B for relevant calculations and justification.**

Our protein is eco-friendly, accelerates drying times by 1.4x, and needs no new equipment:

Fully bio-based (from Trichoderma reesei fungi)

Compatible with existing water-based binders

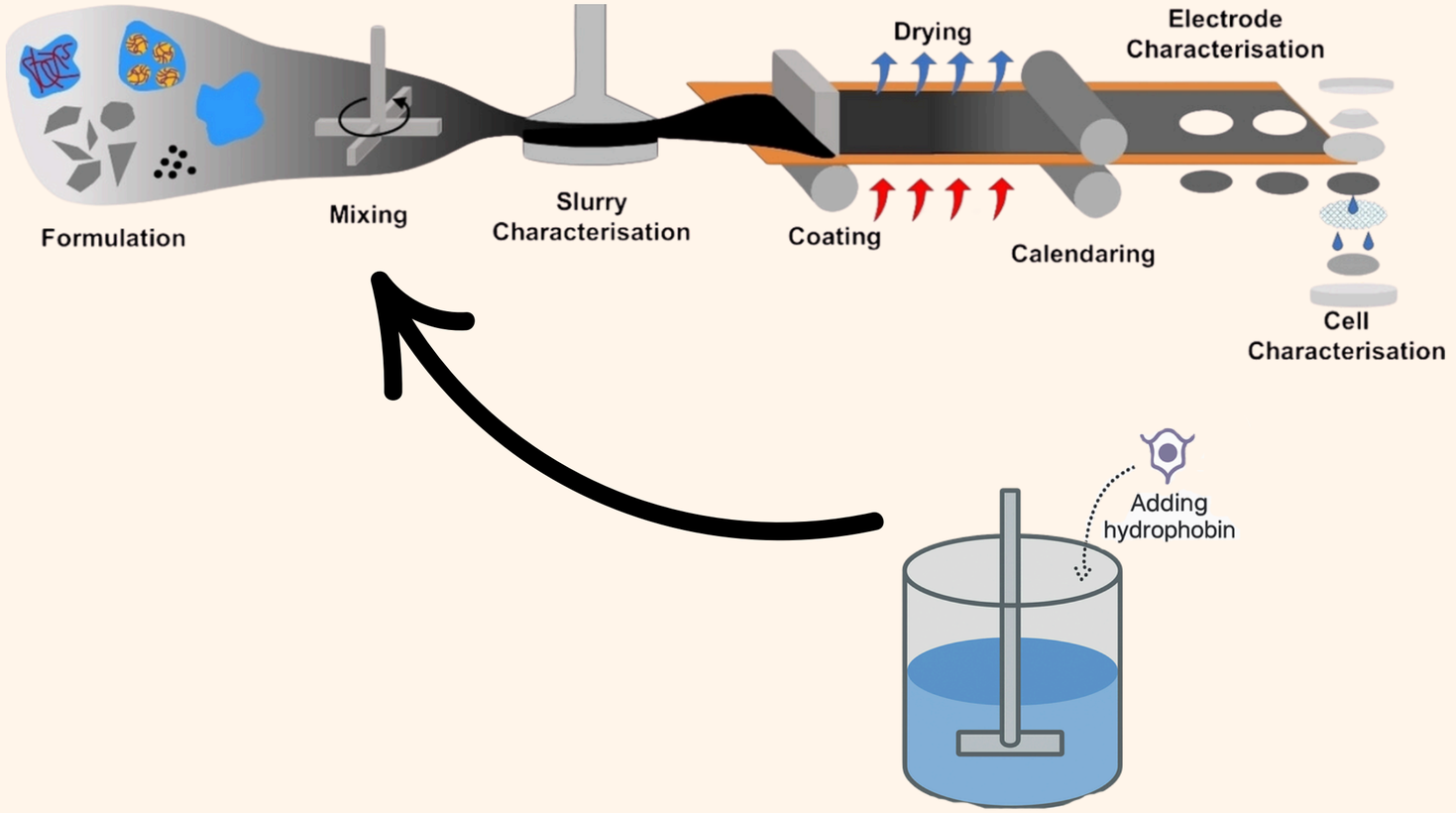
Speeds up drying by 30–50%

Drop-in solution — no equipment or process overhaul required

Supports IP protection & licensing flexibility

*Refer to Appendix B for relevant calculations and justification.

Visualizing the direct addition of Hydrophobin to the Mixing Process of Battery Manufacturing



We will introduce the hydrophobin protein directly into the electrode slurry during the mixing stage of battery manufacturing. This ensures uniform distribution throughout the mixture, effectively reducing drying times in the subsequent processing steps.

Breaking Down The Implementation Plan

Phase 1 – Production:

1. Protein Design & Host Engineering
2. Fermentation & Production at Scale
3. Purification, Formulation & Delivery

Phase 2 – Distribution:

4. Pivot Industries of Focus
5. Inital Development Partnerships
6. IP Rights and Supplying to Clients

Step 1: Designing the Protein for Scalable Production:

1 Chosen Protein

Selected Hydrophobin HFBI, a Class II surface-active protein from Trichoderma reesei, valued for its **water solubility, reversible film formation, and industrial utility.**

2 Host Organism

The gene is expressed in **Pichia pastoris, an industrial yeast** that ensures high protein yields, secretion into the culture medium, and cost-efficient processing.

3 Engineering for Performance

The protein is **codon-optimized** and structurally enhanced for yield, thermal stability, and solubility in water-based binders, **lowering production costs and improving drying performance.**

Step 2: Bioreactor Fermentation & Protein Harvesting:

1 Bioreactor Cultivation

Fermentation is conducted in **50,000 L** stainless steel aerated bioreactors, using oxygenated sugar-rich media and pH control to maximize protein productivity with **batch times of ~48–72 hours.**

2 Scalable Yields

Each batch produces **150–250 kg of hydrophobin**, supplying multiple factories with stable output for months, maintaining **costs at ~$25/gram.**

3 Batch Stability

Fermentation is optimized for consistency and robustness, enabling monthly production cycles that meet partner demand with high purity and activity.

Step 3: From Protein to Factory-Ready Additive

1 Downstream Purification

A three-step purification process—**centrifugation, ultrafiltration, and chromatography**—isolates and concentrates the protein while preserving its function.

2 Binder-Ready Formulation

The protein is **stabilized as a freeze-dried powder or aqueous concentrate** based on partner preference, designed for direct mixing into slurry tanks without altering binder chemistry or pH balance.

3 Seamless Factory Integration

Partners receive **pre-dosed protein packs** that integrate seamlessly into slurry mixing, requiring no process changes while delivering immediate energy savings and faster drying times.

Step 4: Maverick should prioritise industries where cost savings matter more than perfomance:

When compared to PVDF+NMP binders, hydrophobin-enhanced water-based binders have a **5%** performance gap

A 5% performance drop may seem small, but it significantly affects **EV efficiency**. Focusing first on cost-sensitive industries, where this impact is minimal, allows for later EV optimization.

1

IOT & Wearables

2

Consumer Electronics

3

Entry-Level EVs (LFP Powered)

*Refer to Appendix B for relevant calculations and justification.

Step 5: Maverick must partner with battery manufacturers and suppliers for early development of Hydrophobin-aided binders

Partnering allows shared R&D costs to scale the protein from prototype to industry, helping Maverick secure **early capital while retaining IP and equity.**

Collaborating with manufacturers enables Maverick to **validate performance in real binder systems** while maintaining 100% IP through structured tech-transfer and formulation-only agreements.

Companies working in the desired domains:



Through this approach, Maverick is the only one fusing biotech with battery tech, while collaborating with industry giants, instead of competing.

PROFITABILITY ANALYSIS

Key costs that Maverick Biometals needs to incur:

Maverick Biometals drives scalable protein production for cleaner, faster battery manufacturing. At just **$25 per gram**, hydrophobin is cost-effective for industry use. **A $450k one-time R&D investment** secures IP, while **$1.5M funds full production per factory** over a year. This model ensures long-term value through reusable strains, centralized manufacturing, and high-margin protein delivery optimized for large-scale adoption.

$25

to produce 1g of the desired hydrophobin-based protein

$450k

for the original investment into the development of the hydrophobin-based protein to gain IP rights

$1.5M

to produce required amounts of the hydrophobin-based protein per factory for one year.

***Refer to Appendix C for relevant calculations and justification.**

This setup ensures large-scale revenue for Maverick:

Total revenue after supplying the hydrophobin-based protein will be —— **$250M** by 2035

How does this work?

1.

Charging **$4M for supplying the protein** to a client for one factory for three years = $12M

2.

Estimating each client will utilise services for only 4 of their factories to seamlessly phase-in the solution

3.

Estimating 5 total battery partners as forthcoming clients

***Refer to Appendix C for relevant calculations and justification.**

Battery companies can drive upto 1.9x more profit:

(Based on the production of 1GWh of batteries per year.)

CAPEX is saved hereafter through the elimindation of the further requirement and maintainance of NMP safety infrastrucutre.

OPEX requirements also reduce, with a 40% reduction in drying time proportionally resulting in a 40% reducion in energy usage and costs.

For every $1 spent by the battery company on the hydrophobin protein, they save $13 in drying expenditure.

***Refer to Appendix A for relevant calculations and justification.**

$30M
one-time CAPEX savings

$20M
drying OPEX savings

$18.5M
net gains

12x
return on investment

93%
drying efficiency margin

1:13
drying cost-saving ratio

Year	Cost	Revenue	Net Gain
1	**450K**	-	**-450k**
2	**4M**	**5M**	**1M**
3	**6M**	**10M**	**4M**
4	**6M**	**10M**	**4M**
5	**3M**	**5M**	**2M**
6-9	**90M**	**240M**	**150M**

62.5% Profit Margin*

*Based on standard selling price of $4M

*Refer to Appendix C for relevant calculations and justification.

Maverick Biometals can also generate 1.6x more profit:

Key Numbers:

Cost for Producing Protein per Factory per Year = **$1.5M**

Standard Price Charged for Supplying Protein per Factory per Year = **$4M**

Reduced Price Charged to the first partner for Supplying Protein per Factory per Year = **$4M**

Total Factories Protein Being Supplied to per Company = **4**

Number of Battery Partners after 5 Years = **5**

With a strong IP strategy, Maverick can make $20M/year in savings.

Main Business Arguement:

1. Requirement in Context of PVDF and NMP Ban
2. Opportunities for Installment Payments
3. CAPEX and OPEX Savings
4. Leveraging Intellectual Property Rights
5. Potential ESG Benefits

Maverick's $4M annual price reflects **exclusive value, not just cost**. While enzyme production may cost ~$1.5M, partners gain over $20M/year in savings from faster drying, lower energy usage, and no NMP-related capex. The margin ensures continuous innovation, quality, and support.

With full IP control, Maverick is the only provider of this protein. This exclusivity gives partners a strong competitive edge — making $4M a strategic investment into proprietary tech they can't get anywhere else.

By adopting AquaFuse, Maverick can have a sustainable and innovative outcome:

Environmentally Benefical

Eliminates up to 50 tons of NMP per factory annually, preventing the release of ~600,000 kg of CO_2 equivalents each year.

Highly Energy Efficient

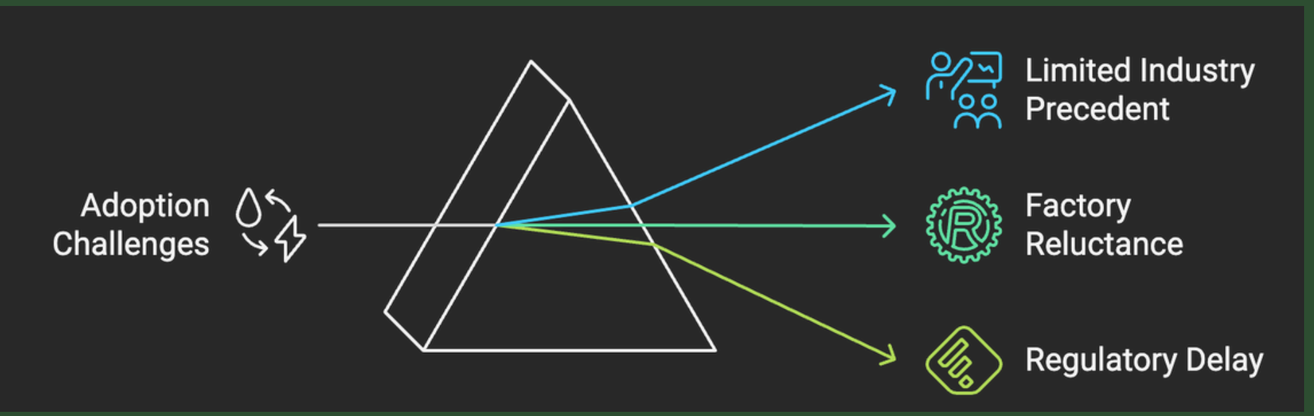
Saves ~5,000–7,000 MWh per factory annually—equivalent to the yearly electricity consumption of 500–700 U.S. homes.

Impactful IP Rights

Opportunity to patent the protein's application for drying water-based binders, potentially **increasing valuation multiples by 20–40%.**

*Refer to Appendix C for relevant calculations and justification.

Risk Analysis



No major precedent for using proteins in electrode drying, making it unproven for skeptics, and needing market education.

Even with cost savings, large factories may resist changes to a core process that already meets yield targets and throughput.

If EU and US enforcement delays, urgency to switch to water-based systems drops — reduces short-term adoption.

Way Forward

Leverage IP to establish **first-mover credibility** while conducting early pilots to generate performance data and publish results for quantified validation.

Impactful IP Rights

Implement a **phased rollout** for seamless integration, enabling real-world validation of cost savings and **performance before full-scale adoption.**

Phase-In Approach

Pivot to regions like the EU and India, targeting consumer electronics brands like Apple, Oppo, and Google, which **prioritize green metrics beyond regulatory requirements.**

Effective Pilot Operations



Nirvaan Duggirala



Aditya Vellanki



Vipran Vasan

On a Personal Note

Dear Maverick Biometals Team,

We have been truly inspired by the opportunity to explore the innovative intersection of biology and energy through the challenge of advancing battery technologies.

We are grateful for the chance to learn from your team's pioneering work and vision, which has been instrumental in shaping our approach to this transformative project. Your commitment to pushing scientific boundaries has fueled our own curiosity and creativity.

As future scientists and innovators, we took this challenge to heart and imagined ourselves contributing to your mission. We're excited about the potential of our ideas to align with your values and contribute meaningfully to your efforts in creating sustainable, biologically driven energy solutions.

We would love the opportunity to continue collaborating and developing these ideas with you. We are eager to help you further this exploration and bring forward solutions that could positively impact the world. Thank you for inspiring and guiding us!

- Nirvaan, Aditya, and Vipran

Source Appendix

A
PROBLEM QUANTIFICATION

B
SOLUTION DESCRIPTION

C
BUSINESS MODEL